Generalized Interpolating Discrete Diffusion (GIDD)

Mathematical Foundations of Discrete Diffusion Models

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Outline

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2. Continuous-Time Markov Chain Construction
3. Forward Transition Rate Matrix R_t
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Motivation: Discrete Diffusion

- Many real-world data are discrete/categorical: text tokens, molecular types, pixel classes
- Continuous diffusion (Gaussian noise) does not preserve categorical structure
- **GIDD** provides a unified framework:
    - Interpolation between data and noise via mixing rates
    - Flexible mixing distributions (mask, uniform, or combinations)
    - Recovers MDM (Masked Diffusion Model) as a special case

Key idea: Define a continuous-time Markov chain (CTMC) on a discrete state space V with $|V|$ categories.

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Mixing Rate

Definition 1.1 (Mixing Rate)

The **cumulative mixing rate** is a differentiable, monotonically decreasing function

$$\alpha : [0,1] \to [0,1], \quad t \mapsto \alpha_t$$

with boundary conditions $\alpha_0 = 1$ (no mixing) and $\alpha_1 = 0$ (fully mixed).

Define $\beta_t := 1 - \alpha_t$. The **signal-to-noise ratio** (SNR) is:

$$\mathrm{SNR}(t) := \frac{\alpha_t}{\beta_t}$$

- At $t = 0$: $\mathrm{SNR} = \infty$ (pure signal)
- At $t = 1$: $\mathrm{SNR} = 0$ (pure noise)

Mixing Distribution

Definition 1.2 (Mixing Distribution)

The **mixing distribution** is a differentiable function

$$\bar{\pi} : [0,1] \to \Delta^{|V|-1}, \quad t \mapsto \bar{\pi}_t$$

where $\Delta^{d-1} = \left\{ \mathbf{x} \in \mathbb{R}^d \,\middle|\, x_i \geq 0,\ \sum_{i=1}^{d} x_i = 1 \right\}$ is the probability simplex.

- $\bar{\pi}_t$ determines the type of noise added at time t
- $\bar{\pi}_1$ is the **prior distribution** of the diffusion process
- **Special case**: $\bar{\pi}_t = \mathbf{m}$ (mask token one-hot) $\Rightarrow$ Masked Diffusion Model

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Marginal Forward Transition

Definition 1.3 (Marginal Forward Transition)

Let $x \in V$ be a data point and $\mathbf{x} \in \{0,1\}^{|V|}$ its one-hot encoding. The **marginal forward transition** is:

$$q_t(z_t|x) := \mathrm{Cat}\left(z_t;\, \alpha_t \mathbf{x} + \beta_t \bar{\pi}_t\right)$$

where $z_t \in V$ is the latent variable at time t.

Interpretation: At time t, with "probability" α_t we keep the data x, and with "probability" β_t we sample from the noise distribution $\bar{\pi}_t$.

- $t = 0$: $q_0(z_0|x) = \mathrm{Cat}(z_0; \mathbf{x}) = \delta_{z_0,x}$ (exact data)
- $t = 1$: $q_1(z_1|x) = \mathrm{Cat}(z_1; \bar{\pi}_1)$ (pure noise)

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Transition Matrix $Q_{t|s}$

Proposition 1.1

There exists a continuous-time Markov chain with transition probability from z_s to z_t $(s \leq t)$:

$$q_{t|s}(z_t|z_s) = \mathrm{Cat}(z_t;\ Q_{t|s}\, z_s)$$

where

$$Q_{t|s} = \alpha_{t|s}\, I + \beta_{t|s}\, \bar{\pi}_{t|s}\, \mathbf{1}^T$$

Here the conditional parameters are:

$$\alpha_{t|s} = \frac{\alpha_t}{\alpha_s}, \qquad \beta_{t|s}\, \bar{\pi}_{t|s} = \beta_t\, \bar{\pi}_t - \frac{\alpha_t}{\alpha_s}\, \beta_s\, \bar{\pi}_s$$

Proof Strategy: Mathematical Induction

Goal: Prove $Q_{t|s} = \alpha_{t|s}\, I + \beta_{t|s}\, \bar{\pi}_{t|s}\, \mathbf{1}^T$ on a discrete grid $\{0, \Delta, 2\Delta, \dots, 1\}$.

Setup: Define single-step quantities

$$\dot{\alpha}_t := \frac{\alpha_{t+\Delta}}{\alpha_t}, \qquad \dot{\beta}_t\, \dot{\bar{\pi}}_t := \beta_{t+\Delta}\, \bar{\pi}_{t+\Delta} - \frac{\alpha_{t+\Delta}}{\alpha_t}\, \beta_t\, \bar{\pi}_t$$

Single-step transition:

$$\dot{Q}_t = \dot{\alpha}_t\, I + \dot{\beta}_t\, \dot{\bar{\pi}}_t\, \mathbf{1}^T$$

Cumulative transition:

$$Q_{t|s} = \prod_{i=s/\Delta}^{t/\Delta-1} \dot{Q}_{\Delta i}$$

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Induction Proof

Base case $(t = s)$:

$$Q_{s|s} = I \quad \checkmark \qquad (\alpha_{s|s} = 1,\ \beta_{s|s}\,\bar{\pi}_{s|s} = 0)$$

Inductive step: Assume $Q_{t|s} = \alpha_{t|s}\,I + \beta_{t|s}\,\bar{\pi}_{t|s}\,\mathbf{1}^T$. Then:

$$Q_{t+\Delta|s} = \dot{Q}_t \cdot Q_{t|s} = \left(\dot{\alpha}_t\,I + \dot{\beta}_t\,\dot{\bar{\pi}}_t\,\mathbf{1}^T\right)\left(\alpha_{t|s}\,I + \beta_{t|s}\,\bar{\pi}_{t|s}\,\mathbf{1}^T\right)$$

Using two key auxiliary lemmas:

- (H1): $\mathbf{1}^T\bar{\pi}_{t|s} = 1$ (probability normalization)
- (H2): $\alpha_{t|s} + \beta_{t|s} = 1$

We obtain: $Q_{t+\Delta|s} = \alpha_{t+\Delta|s}\,I + \beta_{t+\Delta|s}\,\bar{\pi}_{t+\Delta|s}\,\mathbf{1}^T$ $\square$

Induction: Key Computation

Expanding the product:

$$
\begin{aligned}
Q_{t+\Delta|s} &= \dot{\alpha}_t \alpha_{t|s}\, I + \dot{\beta}_t \Big\{ \alpha_{t|s}\, \bar{\pi}_t\, \mathbf{1}^T + \beta_{t|s}\, \bar{\pi}_t\, \underbrace{(\mathbf{1}^T \bar{\pi}_{t|s})}_{=1}\, \mathbf{1}^T \Big\} + \dot{\alpha}_t\, \beta_{t|s}\, \bar{\pi}_{t|s}\, \mathbf{1}^T \\
&= \dot{\alpha}_t \alpha_{t|s}\, I + \dot{\beta}_t\, \underbrace{(\alpha_{t|s} + \beta_{t|s})}_{=1}\, \bar{\pi}_t\, \mathbf{1}^T + \dot{\alpha}_t\, \beta_{t|s}\, \bar{\pi}_{t|s}\, \mathbf{1}^T \\
&= \underbrace{\dot{\alpha}_t \alpha_{t|s}}_{\alpha_{t+\Delta|s}}\, I + \underbrace{\left(\dot{\beta}_t\, \bar{\pi}_t + \dot{\alpha}_t\, \beta_{t|s}\, \bar{\pi}_{t|s}\right)}_{\beta_{t+\Delta|s}\, \bar{\pi}_{t+\Delta|s}}\, \mathbf{1}^T
\end{aligned}
$$

Here we used auxiliary lemmas (H3) and (H4) to verify the identifications.

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From Discrete Grid to Continuous Time

The induction proof holds on the grid $\{0, \Delta, 2\Delta, \dots, 1\}$. To extend to all $s, t \in [0, 1]$:

By differentiability of α_t and $\bar{\pi}_t$, the **rate matrix** exists:

$$R_t = \lim_{\Delta \to 0} \frac{Q_{t+\Delta|t} - I}{\Delta} = \lim_{\Delta \to 0} \frac{\dot{Q}_t - I}{\Delta}$$

Kolmogorov Forward Equation

If R_t is continuous and $\sup_{t \in [0,1]} \|R_t\| < \infty$, then there exists a unique transition matrix family $\{Q_{t|s}\}_{0 \le s \le t \le 1}$ satisfying:

$$\frac{\partial}{\partial t} Q_{t|s} = Q_{t|s} \, R_t, \quad Q_{s|s} = I$$

and the Chapman–Kolmogorov equation: $Q_{t|s} = Q_{t|u} \, Q_{u|s}$ for $s \le u \le t$.

Auxiliary Lemmas

Lemma H1: $\mathbf{1}^T\bar{\pi}_{t|s} = 1$

$$\beta_{t|s}\,\mathbf{1}^T\bar{\pi}_{t|s} = \beta_t - \frac{\alpha_t}{\alpha_s}\,\beta_s$$
$$= \beta_{t|s}$$

Lemma H2: $\alpha_{t|s} + \beta_{t|s} = 1$

$$\frac{\alpha_t}{\alpha_s} + \beta_t - \frac{\alpha_t}{\alpha_s}\beta_s$$
$$= \frac{\alpha_t}{\alpha_s}(1 - \beta_s) + \beta_t = \alpha_t + \beta_t = 1$$

Lemma H3

$\alpha_{t+\Delta|s} = \dot{\alpha}_t\,\alpha_{t|s}$

Proof: $\frac{\alpha_{t+\Delta}}{\alpha_s} = \frac{\alpha_{t+\Delta}}{\alpha_t}\cdot\frac{\alpha_t}{\alpha_s}$

Lemma H4

$\beta_{t+\Delta|s}\,\bar{\pi}_{t+\Delta|s} = \dot{\beta}_t\,\bar{\pi}_t + \dot{\alpha}_t\,\beta_{t|s}\,\bar{\pi}_{t|s}$

Proof: Both sides equal

$\beta_{t+\Delta}\,\bar{\pi}_{t+\Delta} - \frac{\alpha_{t+\Delta}}{\alpha_s}\,\beta_s\,\bar{\pi}_s$

Corollary: Marginal Transition Probability

Corollary 1.2 (Marginal Transition)

The marginal transition probability of the CTMC is:

$$q_t(z_t|x) = \mathrm{Cat}(z_t;\ Q_t\,\mathbf{x}), \qquad Q_t = \alpha_t\, I + \beta_t\, \bar{\pi}_t\, \mathbf{1}^T$$

Proof: Set $s = 0$ in Proposition 1.1. By boundary conditions $\alpha_0 = 1$, $\beta_0 = 0$:

$$\alpha_{t|0} = \alpha_t, \qquad \beta_{t|0}\, \bar{\pi}_{t|0} = \beta_t\, \bar{\pi}_t$$

This gives:

$$q_t(z_t|x) = z_t^T\, Q_t\, \mathbf{x} = \alpha_t\, \delta_{z_t,x} + \beta_t\, (\bar{\pi}_t)_{z_t}$$

which is exactly Definition 1.3. □

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CTMC Forward Transition

Definition 2.1 (CTMC Forward Transition)

For $s < t = s + \Delta$ with $\Delta \to 0$, the forward transition probability satisfies:

$$q_{t|s}(z_t|z_s) = \delta_{z_s,z_t} + R_t(z_s, z_t)\,\Delta + o(\Delta)$$

where R_t is the **forward transition rate matrix**.

Properties of R_t:

- Off-diagonal: $R_t(z, z') \geq 0$ for $z \neq z'$ (instantaneous jump rate)
- Diagonal: $R_t(z, z) = -\sum_{z' \neq z} R_t(z, z')$ (probability conservation)

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GIDD Forward Rate (Lemma 3.6)

Lemma 2.1 (GIDD Forward Transition Rate)

The rate matrix for the GIDD forward process is:

$$R_t(z_s, z_t) = \frac{\alpha_t'}{\alpha_t}\,\delta_{z_s,z_t} + z_t^T\left(\beta_t\,\bar{\pi}_t' - \frac{\alpha_t'}{\alpha_t}\,\bar{\pi}_t\right)$$

Physical interpretation:

- $R_t(z, z')\,dt$ = instantaneous probability of jumping from z to z' in $[t, t+dt)$
- Diagonal: $R_t(z, z) = \frac{\alpha_t'}{\alpha_t} < 0$ (since α_t is decreasing)
- The rate of leaving state z equals $-\frac{\alpha_t'}{\alpha_t} > 0$

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Proof of Lemma 2.1

Taylor expand $q_{s+\Delta|s}(z_t|z_s)$ at $\Delta = 0$:

$$\alpha_{s+\Delta|s} = \frac{\alpha_{s+\Delta}}{\alpha_s} = 1 + \frac{\alpha'_s}{\alpha_s}\Delta + o(\Delta)$$

$$\begin{aligned} \beta_{s+\Delta|s}\,\bar{\pi}_{s+\Delta|s} &= \beta_{s+\Delta}\,\bar{\pi}_{s+\Delta} - \frac{\alpha_{s+\Delta}}{\alpha_s}\beta_s\,\bar{\pi}_s \\ &= \underbrace{\left[(\beta_s\,\bar{\pi}_s)' - \frac{\alpha'_s}{\alpha_s}\beta_s\,\bar{\pi}_s\right]}_{\text{simplifies to } \beta_s\,\bar{\pi}'_s - \frac{\alpha'_s}{\alpha_s}\bar{\pi}_s}\Delta + o(\Delta) \end{aligned}$$

Substituting:

$$\begin{aligned} q_{s+\Delta|s}(z_t|z_s) &= \delta_{z_s,z_t} \\ &\quad + \underbrace{\left[\frac{\alpha'_s}{\alpha_s}\delta_{z_s,z_t} + z_t^T\left(\beta_s\,\bar{\pi}'_s - \frac{\alpha'_s}{\alpha_s}\bar{\pi}_s\right)\right]}_{=R_s(z_s,z_t)}\Delta + o(\Delta) \end{aligned}$$

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Simplification of the Rate Expression

The key simplification uses the product rule. Starting from:

$$(\beta_s \bar{\pi}_s)' - \frac{\alpha_s'}{\alpha_s} \beta_s \bar{\pi}_s$$

Expand with $\beta_s = 1 - \alpha_s$:

$$\begin{aligned}
&= (1-\alpha_s)' \bar{\pi}_s + \beta_s \bar{\pi}_s' - \frac{\alpha_s'}{\alpha_s} (1-\alpha_s) \bar{\pi}_s \\
&= -\alpha_s' \bar{\pi}_s + \beta_s \bar{\pi}_s' - \frac{\alpha_s'}{\alpha_s} \beta_s \bar{\pi}_s \\
&= \beta_s \bar{\pi}_s' - \alpha_s' \left(1 + \frac{\beta_s}{\alpha_s}\right) \bar{\pi}_s = \beta_s \bar{\pi}_s' - \frac{\alpha_s'}{\alpha_s} \bar{\pi}_s
\end{aligned}$$

This gives the clean expression: $R_t = \frac{\alpha_t'}{\alpha_t} I + \left(\beta_t \bar{\pi}_t' - \frac{\alpha_t'}{\alpha_t} \bar{\pi}_t\right) \mathbf{1}^T$

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Backward Transition: Setup

Goal: Derive the backward rate $\hat{R}_t$ for denoising (recovering data from noise).

Definition 3.1 (CTMC Backward Transition)

For $s = t - \Delta$, $\Delta \to 0$:

$$p_{s|t}(z_s|z_t) = \delta_{z_t, z_s} + \hat{R}_t(z_t, z_s)\,\Delta + o(\Delta)$$

Lemma 3.1 (Bayesian Form of Backward Transition)

The true backward transition is:

$$q_{s|t}(z_s|z_t, x) = \frac{q_{t|s}(z_t|z_s) \cdot q_s(z_s|x)}{q_t(z_t|x)}$$

This follows from Bayes' rule and the Markov property: $q_{t|s}(z_t|z_s, x) = q_{t|s}(z_t|z_s)$.

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Model Backward Distribution

Since we don't know the true x, we use a neural network $x_\theta(Z_t, t)$ to predict it.

Model backward distribution:

$$p_\theta(z_s|z_t) := q_{t|s}(z_t|z_s) \cdot \frac{q_s(z_s|x_\theta)}{q_t(z_t|x_\theta)}$$

where $q_t(z_t|x_\theta) := \mathrm{Cat}(z_t;\ Q_t\, x_\theta(Z_t, t))$.

Key idea: Replace the unknown true x with the model's prediction $x_\theta(Z_t, t)$, preserving the same Bayesian structure.

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GIDD Backward Rate

Lemma 3.2 (GIDD Backward Rate)

The CTMC backward rate for the model posterior is:

$$\hat{R}_t^\theta(z_t, z_s) = -\delta_{z_s, z_t} \sum_{z'} R_t(z', z_t) \frac{q_t(z'|x_\theta)}{q_t(z_t|x_\theta)} + R_t(z_s, z_t) \frac{q_t(z_s|x_\theta)}{q_t(z_t|x_\theta)}$$

Equivalently, in cases:

$$\hat{R}_t^\theta(z_t, z_s) = \begin{cases} R_t(z_s, z_t) \dfrac{q_t(z_s|x_\theta)}{q_t(z_t|x_\theta)} & \text{if } z_s \neq z_t \\ -\displaystyle\sum_{z' \neq z_t} R_t(z', z_t) \dfrac{q_t(z'|x_\theta)}{q_t(z_t|x_\theta)} & \text{if } z_s = z_t \end{cases}$$

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Proof of Backward Rate (1/2)

Set $s = t - \Delta$, $\Delta \to 0$. From the model backward distribution:

$$p_\theta(z_s|z_t) = q_{t|s}(z_t|z_s) \cdot \frac{q_s(z_s|x_\theta)}{q_t(z_t|x_\theta)}$$

Substitute the forward rate expansion and Taylor expand q_s:

$$p_\theta(z_s|z_t) = \left(\delta_{z_s,z_t} + R_t(z_s, z_t)\,\Delta + o(\Delta)\right) \cdot \frac{q_t(z_s|x_\theta) - q_t'(z_s|x_\theta)\,\Delta + o(\Delta)}{q_t(z_t|x_\theta)}$$

Proof of Backward Rate (2/2)

Expanding to first order in Δ:

$$p_\theta(z_s|z_t) = \delta_{z_s,z_t} + \left[-\delta_{z_s,z_t}\frac{q_t'(z_s|x_\theta)}{q_t(z_t|x_\theta)} + R_t(z_s,z_t)\frac{q_t(z_s|x_\theta)}{q_t(z_t|x_\theta)}\right]\Delta + o(\Delta)$$

Comparing with the definition $p_{s|t}(z_s|z_t) = \delta_{z_t,z_s} + \hat{R}_t(z_t,z_s)\,\Delta + o(\Delta)$:

$$\hat{R}_t^\theta(z_t,z_s) = -\delta_{z_s,z_t}\frac{q_t'(z_t|x_\theta)}{q_t(z_t|x_\theta)} + R_t(z_s,z_t)\frac{q_t(z_s|x_\theta)}{q_t(z_t|x_\theta)}$$

Finally, apply the **Kolmogorov forward equation**:

$$q_t'(z|x_\theta) = \sum_{z'} q_t(z'|x_\theta)\,R_t(z',z)$$

to eliminate the time derivative and obtain the final form.

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General CT-ELBO

Proposition 4.1 (General CT-ELBO, Proposition H.4)

For any CTMC diffusion process with marginal q_t, forward rate R_t, and backward rate $\hat{R}_t$:

$$\log p(x) \geq \mathbb{E}_{t, q_t(z_t|x)} \left[\hat{R}_t(z_t, z_t) - R_t(z_t, z_t) + \sum_{z' \neq z_t} R_t(z', z_t) \frac{q_t(z'|x)}{q_t(z_t|x)} \log \frac{\hat{R}_t(z_t, z') \, q_t(z_t|x)}{R_t(z', z_t) \, q_t(z'|x)} \right] + C$$

where $t \sim \mathcal{U}(0,1)$ and

$$C = \mathbb{E}_{q_0(z_0|x)}[\log p(x|z_0)] - D_{\mathrm{KL}}(q_1(z_1|x) \| p_1(z_1))$$

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Proof Sketch of CT-ELBO (1/3)

Step 1: Start from the KL divergence between true and model backward:

$$D_{\mathrm{KL}}(q_{s|t}(\cdot|z_t, x) \| p_{s|t}(\cdot|z_t)) = \sum_{z_s} \frac{q_{t|s}(z_t|z_s)\, q_s(z_s|x)}{q_t(z_t|x)} \log \frac{q_{t|s}(z_t|z_s)\, q_s(z_s|x)}{p_{s|t}(z_s|z_t)\, q_t(z_t|x)}$$

Step 2: Taylor expand each factor for $s = t - \Delta$, $\Delta \to 0$:

- For $z_s = z_t$: $\log q_{t|s}(z_t|z_s) = \Delta\, R_s(z_t, z_t) + o(\Delta)$
- For $z_s \neq z_t$: the product $q_{t|s} \log q_{t|s}$ contributes at order $O(\Delta)$

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Proof Sketch of CT-ELBO (2/3)

Step 2 (continued): The KL divergence expands as:

$$D_{\mathrm{KL}}(q_{s|t}(\cdot|z_t, x) \| p_{s|t}(\cdot|z_t)) = \Delta\Big(R_s(z_t, z_t) - \hat{R}_s(z_t, z_t) + \sum_{z_s \neq z_t} R_s(z_s, z_t) \frac{q_s(z_s|x)}{q_t(z_t|x)} \log \frac{R_s(z_s, z_t)\, q_s(z_s|x)}{\hat{R}_s(z_t, z_s)\, q_t(z_t|x)}\Big) + o(\Delta)$$

Key observation: The KL divergence is $O(\Delta)$, not $O(1)$.

Proof Sketch of CT-ELBO (3/3)

Step 3: From discrete to continuous. The discrete-time ELBO is:

$$\log p(x) \geq -\sum_{i=2}^{T} \mathbb{E}_{q_{\Delta i}(z_t|x)} \left[D_{\mathrm{KL}}(q_{\Delta(i-1)|\Delta i}(\cdot|z_t, x) \| p_\theta(\cdot|z_t)) \right] + C$$

With $\Delta = 1/T$ and using the $O(\Delta)$ expansion:

$$\log p(x) \geq -\frac{1}{\Delta} \cdot \mathbb{E}_{t \sim \mathcal{U}(0,1)} [\Delta\,(\text{rate terms}) + o(\Delta)] + C$$

The Δ's cancel, $o(\Delta)/\Delta \to 0$, yielding the CT-ELBO. $\square$

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Weight Function

Lemma 4.1 (Weight Expectation, Lemma H.5)

Define the weight function:

$$w_t(z_t, x) = \frac{1}{q_t(z_t|x)}\, z_t^T \left(\beta_t\, \bar{\pi}_t' - \frac{\alpha_t'}{\alpha_t}\, \bar{\pi}_t\right)$$

Then:

$$\mathbb{E}_{z_t \sim q_t(\cdot|x)}[w_t(z_t, x)] = -\frac{\alpha_t'}{\alpha_t}$$

Proof: Since $\sum_{z_t} z_t = \mathbf{1}$ (summing over all one-hot vectors):

$$\mathbb{E}_{z_t}[w_t] = \mathbf{1}^T \left(\beta_t\, \bar{\pi}_t' - \frac{\alpha_t'}{\alpha_t}\, \bar{\pi}_t\right) = \beta_t \underbrace{(\mathbf{1}^T \bar{\pi}_t')}_{=(\mathbf{1}^T \bar{\pi}_t)'=0} - \frac{\alpha_t'}{\alpha_t} \underbrace{(\mathbf{1}^T \bar{\pi}_t)}_{=1} = -\frac{\alpha_t'}{\alpha_t}$$

Key: $\bar{\pi}_t$ is always a probability vector, so $\mathbf{1}^T \bar{\pi}_t = 1 \Rightarrow \mathbf{1}^T \bar{\pi}_t' = 0$.

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GIDD ELBO Theorem

Theorem 4.1 (GIDD ELBO, Theorem 3.7)

The continuous-time negative ELBO (CT-NELBO) satisfies:

$$-\log p(x) \leq \mathbb{E}_{t,z_t}[w_t(z_t,x)\,(D_{\mathrm{KL}}(q_t(\cdot|x)\|q_t(\cdot|x_\theta)) + D_{\mathrm{IS}}(q_t(z_t|x)\|q_t(z_t|x_\theta)))] + C$$

where $t \sim \mathcal{U}(0,1)$, $z_t \sim q_t(\cdot|x)$, and

$$D_{\mathrm{IS}}(p\|q) = \frac{p}{q} - \log\frac{p}{q} - 1$$

is the pointwise **Itakura–Saito divergence**.

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GIDD ELBO: Proof Outline (1/2)

Step 1: Substitute the GIDD backward rate into the general CT-ELBO.
Step 2: Simplify the first summation term. Using $R_t(z', z_t) = w_t(z_t, x)\, q_t(z_t|x)$ for $z' \neq z_t$:

$$\sum_{z'} R_t(z', z_t) \frac{q_t(z'|x_\theta)}{q_t(z_t|x_\theta)} = w_t(z_t, x) \frac{q_t(z_t|x)}{q_t(z_t|x_\theta)} + \frac{\alpha_t'}{\alpha_t}$$

Step 3: Simplify the logarithmic term:

$$\sum_{z' \neq z_t} \frac{q_t(z'|x)}{q_t(z_t|x)} R_t(z', z_t) \log \frac{q_t(z'|x_\theta)\, q_t(z_t|x)}{q_t(z_t|x_\theta)\, q_t(z'|x)}$$

$$= w_t(z_t, x) \left(-D_{\mathrm{KL}}(q_t(\cdot|x) \| q_t(\cdot|x_\theta)) + \log \frac{q_t(z_t|x)}{q_t(z_t|x_\theta)} \right)$$

Step 4: Combine and use Lemma 4.1.
After merging the two terms:

$$-\log p(x) \leq \mathbb{E}_{t,z_t}\left[w_t\left(D_{\mathrm{KL}} + \frac{q_t(z_t|x)}{q_t(z_t|x_\theta)} - \log\frac{q_t(z_t|x)}{q_t(z_t|x_\theta)}\right) + \frac{\alpha'_t}{\alpha_t}\right] + C$$

Since $\mathbb{E}_{z_t}[w_t(z_t,x)] = -\frac{\alpha'_t}{\alpha_t}$, the constant term $\frac{\alpha'_t}{\alpha_t}$ is absorbed:

$$-\log p(x) \leq \mathbb{E}_{t,z_t}\left[w_t\left(D_{\mathrm{KL}} + \underbrace{\frac{q_t(z_t|x)}{q_t(z_t|x_\theta)} - \log\frac{q_t(z_t|x)}{q_t(z_t|x_\theta)} - 1}_{=D_{\mathrm{IS}}(q_t(z_t|x)\|q_t(z_t|x_\theta))}\right)\right] + C$$

Outline

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Mixing Schedule: Masking + Uniform Noise

Goal: Design a concrete mixing schedule combining mask and uniform noise.

Define the schedule with normalization $C_t = 1 + c_t$:

$$\alpha_t = \frac{1-t}{C_t}, \qquad \beta_t\,\pi_t = \frac{t}{C_t}\,\mathbf{m} + \frac{c_t}{C_t}\,\mathbf{u}$$

where:

- $\mathbf{m}$ = one-hot vector for the mask token
- $\mathbf{u}$ = uniform distribution vector ($1/|V|$ for each entry)
- $c_t = B\,t^{\gamma/2}\,(1-t)^{\gamma/2}$ controls the uniform noise level

Interpretation: At each time t, the noisy distribution is a mixture of data, mask, and uniform noise.

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Uniform Noise Ratio (Lemma H.7)

Lemma 6.1 (Uniform Noise Ratio)

If $c_t = B\, t^{\gamma/2}\,(1-t)^{\gamma/2}$ with $B = 2^\gamma \frac{p_u}{1-p_u}$, then the maximum uniform noise ratio is p_u, achieved at $t = 1/2$.

Proof: The uniform noise ratio is $\frac{c_t}{1+c_t}$.
Since $t(1-t)$ is symmetric and maximized at $t = 1/2$:

$$c_{1/2} = B\,(1/2)^\gamma = 2^\gamma \cdot \frac{p_u}{1-p_u} \cdot 2^{-\gamma} = \frac{p_u}{1-p_u}$$

Maximum ratio:

$$\frac{c_{1/2}}{1+c_{1/2}} = \frac{p_u/(1-p_u)}{1+p_u/(1-p_u)} = \frac{p_u}{1} = p_u \qquad \square$$

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ELBO Weight Function (Lemma H.8)

Lemma 6.2 (GIDD ELBO Weights)

Under this mixing schedule, the weight function has closed form:

$$w_t(z_t, x) = z_t^T \left(\frac{\mathbf{m} + (c_t + (1-t)\, c_t')\, \mathbf{u}}{(1-t)\big((1-t)\, \mathbf{x} + t\, \mathbf{m} + c_t\, \mathbf{u}\big)} \right)$$

Key derivations:

- $c_t' = \frac{\gamma}{2} \cdot \frac{1-2t}{t(1-t)} \cdot c_t$
- $\frac{\alpha_t'}{\alpha_t} = -\frac{1}{1-t} - \frac{c_t'}{1+c_t}$
- Core ratio: $\frac{\beta_t\, \pi_t}{\alpha_t} = \frac{t\, \mathbf{m} + c_t\, \mathbf{u}}{1-t}$
- Its derivative: $\left(\frac{\beta_t\, \pi_t}{\alpha_t}\right)' = \frac{\mathbf{m} + (c_t + (1-t)\, c_t')\, \mathbf{u}}{(1-t)^2}$

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Outline

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Summary of GIDD Framework

1. Forward process: Interpolation $q_t(z_t|x) = \mathrm{Cat}(z_t;\ \alpha_t\,\mathbf{x} + \beta_t\,\bar{\pi}_t)$
2. CTMC construction: Transition matrix $Q_{t|s} = \alpha_{t|s}\,I + \beta_{t|s}\,\bar{\pi}_{t|s}\,\mathbf{1}^{T}$
3. Forward rate: $R_t(z_s, z_t) = \frac{\alpha_t'}{\alpha_t}\,\delta_{z_s,z_t} + z_t^{T}\left(\beta_t\,\bar{\pi}_t' - \frac{\alpha_t'}{\alpha_t}\,\bar{\pi}_t\right)$
4. Backward rate: $\hat{R}_t^{\theta}(z_t, z_s) = R_t(z_s, z_t)\,\frac{q_t(z_s|x_\theta)}{q_t(z_t|x_\theta)}$ (for $z_s \neq z_t$)
5. ELBO: $-\log p(x) \leq \mathbb{E}_{t,z_t}\big[w_t\big(D_{\mathrm{KL}} + D_{\mathrm{IS}}\big)\big] + C$
6. Mixing schedule: Mask + uniform noise with tunable ratio p_u

Key Takeaways

- GIDD **generalizes** existing discrete diffusion models (MDM is a special case with $\bar{\pi}_t = \mathbf{m}$)
- The framework supports **flexible noise types**: mask, uniform, or any mixture
- The ELBO decomposes into a **KL divergence** (global distribution matching) and an **Itakura–Saito divergence** (pointwise likelihood ratio matching)
- The weight function w_t is **independent** of the specific mixing distribution $\bar{\pi}_t$ in expectation (Lemma 4.1)
- All derivations are **rigorous**: discrete grid induction $\rightarrow$ continuous-time limit via Kolmogorov equations

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. How does the choice of mixing distribution $\bar{\pi}_t$ affect generation quality for different data modalities?
2. What is the optimal balance between mask noise and uniform noise (choice of p_u and γ)?
3. How does GIDD compare to autoregressive models for text generation in practice?
4. Can the CTMC framework be extended to structured discrete spaces (e.g., graphs, trees)?

Welcome inputs

Any comments?

Welcome your inputs!